

Express Liner Website

Client, Admin & Developer Manual

Official project handover document

Public website: <https://www.expresslinerco.com/>

Prepared from the ASP.NET Core source code and verified Hostinger VPS deployment

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1. Website Overview

Express Liner is a freight transportation website for presenting the company, services, business information and contact channels. Visitors can request a quote or send a contact message. Administrators manage the content and incoming requests through a protected CMS.

Public routes

Route	Purpose
/	Homepage with hero, about, services, features, contact information and forms.
/Home/Privacy	Privacy page.
/health	JSON health check: { status: healthy }.
/sitemap.xml	XML sitemap containing the public home URL.

Public workflow

A visitor opens the homepage, reviews company information, chooses a service or contact section, completes a form, passes browser/server validation and receives a success notification after the record is saved. Quote and contact records are visible in the admin area; SMTP notification is attempted when production SMTP variables are configured.

Responsive and visual testing

The repository contains Playwright tests for mobile (375px), tablet (768px) and desktop (1440px) layouts, navigation behavior and visual baselines. The visual baseline suite covers the public homepage at all three sizes.

2. Client Manual

Open <https://www.expresslinerco.com/> in a modern browser. Use the top navigation to move to Home, About Us, Services, Why Choose Us and Contact Us. On mobile, use the menu button in the header.

Requesting a quote

- Open Get a Quote or scroll to the quote section.
- Complete name, email, phone, pickup location, delivery location and freight type. Optional fields include company, weight, pickup date and additional information.
- Submit the form. The application validates the fields and saves a new request with status New.
- The business team can then review the request in Admin → Quotes.

Contacting the company

- Open Contact Us and enter name, email and message. Phone is optional.
- Submit the message. It is stored in the admin database and an email notification is attempted when SMTP is configured.

Basic troubleshooting

- Refresh with Ctrl+F5 if an old style or image is displayed.
- If a form does not submit, check required fields and email/phone formats.
- If the website is unavailable, contact the site administrator; the developer can check /health and the systemd logs.

3. Admin Manual

Admin login: <https://expresslinerc.com/Identity/Account/Login>. Authentication uses ASP.NET Core Identity with an Admin role and a CAPTCHA check on the login page. Credentials are managed through production environment variables and are intentionally not printed in this document.

Admin navigation

Menu	Route	Function
Dashboard	/Admin/Dashboard	Summary counters and recent requests/messages.
Quotes	/Admin/QuoteRequests	Search, filter, inspect, change status and delete quote requests.
Messages	/Admin/ContactMessages	Search and delete contact messages.
Services	/Admin/Services	Create, edit, upload image and delete services.
Features	/Admin/Features	Create, edit and delete Why Choose Us features.
Statistics	/Admin/Statistics	Create, edit and delete statistics.
Settings	/Admin/Settings	Edit site, hero, about, contact and SEO data; upload images.

Logout

Use Logout in the admin sidebar. The authentication cookie is cleared and protected routes redirect to the login page.

4. Complete CRUD Reference

Services

Create: Admin → Services → Add Service. Required fields are Title and Short Description. Full Description, Icon, Image URL, Display Order, Active and an optional image upload are available. Save redirects to the list. Edit opens the existing values; an uploaded replacement image is validated and the old upload is deleted. Delete asks for confirmation, removes the database record and deletes its uploaded image.

Features

Create and edit through Admin → Features. Title and Description are required. Icon, Display Order and Active are editable. Save returns to the list; Delete is permanent after confirmation.

Statistics

Create and edit through Admin → Statistics. Value and Label are required. Icon, Display Order and Active are editable. Save returns to the list; Delete is permanent after confirmation.

Settings

Admin → Settings is a single save form. It manages Site Settings, Hero, About, Contact and SEO fields. Logo, hero, about and Open Graph images can be uploaded as JPEG, PNG or WebP. The application limits images to 5 MB. Existing uploaded images are deleted when replaced. Save redirects with a success message.

Quote requests

Admin → Quotes supports search by name, company, email or phone; status filtering; Details; status changes (New, InProgress, Contacted, Completed, Archived); and permanent deletion after confirmation.

Contact messages

Admin → Messages supports search by sender, email, phone or message and permanent deletion after confirmation.

5. Developer Manual

Technology

Item	Actual value
Framework	ASP.NET Core 8 MVC + Razor Pages
Target	.NET 8 (net8.0)
ORM	Entity Framework Core 8
Database provider	Microsoft.EntityFrameworkCore.SqlServer
Authentication	ASP.NET Core Identity + Admin role + cookie authentication
Web server	Kestrel behind Nginx on Hostinger VPS
Email	System.Net.Mail SMTP sender
CAPTCHA	Google reCAPTCHA verifier on Identity login
Testing	Playwright TypeScript in E2E/

Startup flow

Program.cs registers SQL Server DbContext, Data Protection, upload options, SMTP, CAPTCHA, Identity, MVC and Razor Pages. SeedData.InitializeAsync runs migrations, creates the Admin role/user when configuration is present, and seeds website content. Middleware then handles forwarded headers, HTTPS, static files, routing, authentication and authorization.

Important services

Path	Purpose
Services/FileService.cs	Validates and stores image uploads; allows jpg/jpeg/png/webp up to 5 MB.
Services/SmtpEmailSender.cs	Sends quote/contact notifications and sets Reply-To.
Services/GoogleRecaptchaVerifier.cs	Posts the login CAPTCHA token to Google siteverify.
Data/ApplicationDbContext.cs	Identity DbContext plus website DbSet and model configuration.
Data/SeedData.cs	Migrations, Admin role/user and initial content seeding.

6. Project Structure

/Controllers

/Areas/Admin/Controllers

/Areas/Admin/Views

/Areas/Identity/Pages

/Models

/ViewModels

/Data

/Services

/Views

/wwwroot

/E2E

Program.cs

appsettings*.json

Public MVC controllers

Protected admin controllers

Admin Razor views

Identity login and account pages

Domain and Identity models

Input and dashboard models

DbContext, seed and migrations

Upload, SMTP and CAPTCHA services

Public Razor views and layouts

CSS, JavaScript, images and uploads

Playwright TypeScript tests

Application startup and middleware

Configuration templates

Do not edit the template/ project copy for production; the csproj excludes template/** from compilation and content.

7. Database Documentation

The project uses SQL Server. ApplicationDbContext inherits IdentityDbContext. The production VPS uses SQL Server 2022 in Docker container sql1 with database TruckCompany. The application runs Database.MigrateAsync() at startup using the compiled InitialSqlServer migration.

Main entities

Entity/table	Purpose
AspNetUsers / AspNetRoles / AspNetUserRoles	Identity users and Admin role.
SiteSettings	Company, branding and SEO.
HeroSections / AboutSections / ContactSettings	Homepage content.
Services / Features / Statistics / SocialLinks	Public content collections.
QuoteRequests	Public quote submissions and status.
ContactMessages	Public contact submissions.
__EFMigrationsHistory	Applied EF Core migrations.

Migration commands

```
dotnet ef migrations add MigrationName
dotnet ef database update
dotnet ef migrations list
```

Before production migrations: back up the database, test locally, publish the matching application, apply the migration, inspect logs and test the affected workflow. Production connection values are stored only in /etc/truckcompany.env on the VPS.

8. Local Development

Requirements: .NET SDK 8, SQL Server/LocalDB, Git and optionally Node.js for E2E tests. The local development connection is configured in appsettings.Development.json or User Secrets.

```
git clone https://github.com/bakirmaksumov/Truck_company.git
cd Truck_company
dotnet restore
dotnet build
```

```
dotnet ef database update
dotnet run
```

Typical local URLs are defined in Properties/launchSettings.json: <http://localhost:5031> and <https://localhost:7041>. The admin login is /Identity/Account/Login and the admin area is /Admin. Run E2E tests with `npm.cmd install` in E2E, then `npm.cmd test`. Use a dedicated test database, never production.

9. Production Configuration

The repository contains configuration templates only. Production secrets are not committed. On the VPS they are stored in `/etc/truckcompany.env` with mode 600.

Variable	Purpose
ConnectionStrings__DefaultConnection	SQL Server connection to TruckCompany.
Admin__Email / Admin__Password	Initial admin seed credentials.
Smtp__Host/Port/EnableSsl/UserName/Password	Hostinger SMTP delivery.
Smtp__FromEmail / Smtp__ToEmail	Notification sender and recipient.
GoogleRecaptcha__SiteKey / SecretKey	CAPTCHA on admin login.
Uploads__PhysicalPath / RequestPath	Persistent upload storage and URL path.
ASPNETCORE__ENVIRONMENT / ASPNETCORE__URLS	Production mode and local Kestrel bind.

Never commit connection strings, passwords, SMTP credentials, CAPTCHA secret keys, publish profiles or server environment files.

10. Hostinger Deployment Manual

Verified architecture

Hostinger VPS: Ubuntu 24.04.4 LTS at 2.25.242.114. .NET SDK/runtime 8.0.131, SQL Server 2022 Docker container sql1, Kestrel on 127.0.0.1:5000, Nginx on ports 80/443, systemd service truckcompany.service, Let's Encrypt certificate for expresslinerco.com and www.expresslinerco.com.

Verified locations

Item	Value
Application source	<code>/var/www/truckcompany</code>
Published app	<code>/var/www/truckcompany/publish</code>
Persistent uploads	<code>/var/www/truckcompany/uploads</code>
Environment file	<code>/etc/truckcompany.env</code>
systemd unit	<code>/etc/systemd/system/truckcompany.service</code>
Nginx site	<code>/etc/nginx/sites-enabled/truckcompany</code>
SQL container	<code>sql1 / localhost:1433</code>
Health check	<code>http://127.0.0.1:5000/health</code>

Update deployment

```
cd /var/www/truckcompany
git pull origin main
dotnet publish Truck_company.csproj -c Release -o /var/www/truckcompany/publish
systemctl restart truckcompany.service
systemctl status truckcompany.service
curl -I https://expresslinerco.com
```

The current deployment was validated with HTTPS 200 OK and HTTP 301 redirect to HTTPS. DNS A records point @ to 2.25.242.114 and www is a CNAME to expresslinerco.com. DNS values should be rechecked in Hostinger before future migrations.

11. Backup, Update and Rollback

Before every deployment back up the SQL database, application environment file and upload directory. Use Docker volume backup or SQL Server backup tooling appropriate to the production policy. [REQUIRES HOSTINGER SERVER VERIFICATION: exact backup retention and off-server storage policy.]

```
# inspect service and logs
systemctl status truckcompany.service
journalctl -u truckcompany.service -n 100 --no-pager

# inspect database container
docker ps
docker logs --tail 100 sql1

# validate proxy
nginx -t
```

Rollback: stop the service, restore the previous publish directory and environment file, restart the service, inspect logs and test HTTPS, homepage, forms and admin login. Database rollback after a schema migration must be planned separately; do not restore an old database blindly against a newer model.

12. Troubleshooting

Symptom	Diagnosis and action
HTTP 500	journalctl -u truckcompany.service; verify environment variables and database connectivity.
Service inactive	systemctl status truckcompany.service; run systemctl restart and inspect logs.
Database failure	docker ps; docker logs sql1; verify ConnectionStrings__DefaultConnection.
Admin login fails	Check Admin variables, CAPTCHA keys, HTTPS and Identity seed logs.
Email not received	Verify SMTP host/port/user/password; inspect application logs; test the Hostinger mailbox.
Images missing	Check /var/www/truckcompany/uploads, Uploads variables and permissions.
CSS/JS missing	Check Nginx proxy, wwwroot in publish and browser cache.
SSL/DNS issue	Check DNS A/CNAME records, certbot certificates and nginx -t.
Migration fails	Back up first; inspect SQL container and EF migration history.

13. Security and Maintenance

Current protections include Identity password hashing, Admin-role authorization, antiforgery tokens on POST forms, parameterized EF Core queries, HTML encoding in notification bodies, image extension/MIME/size checks, HTTPS, protected environment file permissions and CAPTCHA on login.

Recommended improvements

- Create a non-root deployment user and run the application under it; the current verified service uses root and should be hardened during a planned maintenance window.
- Do not expose SQL Server port 1433 publicly; bind it to localhost or a private Docker network and allow only required firewall traffic.
- Configure Data Protection key encryption or a protected persistent key strategy.
- Rotate any credentials that have ever been shared in chat or stored in old production files.
- Add off-server encrypted backups and a tested restore procedure.

Maintenance checklist

- Check <https://expresslinerco.com/> and [/health](https://expresslinerco.com/health).
- Test quote and contact forms and confirm admin records/email delivery.
- Review systemd and Nginx logs.
- Check SQL container health, disk space, uploads and SSL renewal.
- Run dependency/security updates during a maintenance window.
- Run the Playwright suite before and after production changes.

Quick Reference

Item	Value
Public website	https://www.expresslinerco.com/
Admin URL	https://expresslinerco.com/Identity/Account/Login
Technology	ASP.NET Core 8 MVC + Razor Pages
Database	SQL Server 2022 in Docker container sql1
Hosting	Hostinger VPS / Ubuntu 24.04.4
Service	truckcompany.service
Nginx	<code>/etc/nginx/sites-enabled/truckcompany</code>
Uploads	<code>/var/www/truckcompany/uploads</code>
Publish command	<code>dotnet publish Truck_company.csproj -c Release -o /var/www/truckcompany/publish</code>
Health	https://expresslinerco.com/health

Confidentiality note: passwords, SMTP credentials, CAPTCHA secret keys and database connection strings are intentionally omitted. Retrieve them from the secured VPS environment or Hostinger account, never from Git history.